

Phase 4: ground the product, then assign an arm

Connect an assembly requirement to observed facts and a supported assignment.



10-minute technical walkthrough • PA = ProductAgent • RA = RobotAgent

Phase 4 builds a grounded product context

4.0

Ontology foundation

Supplied PPR TBox
Interaction ABox
Workcell authorities

4.1

Document evidence

Approved page index
PA-authored queries
Cited interpretations

4.2

CAD + RGB-D

Observed candidates
Morphology + size
Calibrated locations

4.3

Evidence contracts

Typed records
Exact references
Hashes + uncertainty

4.4

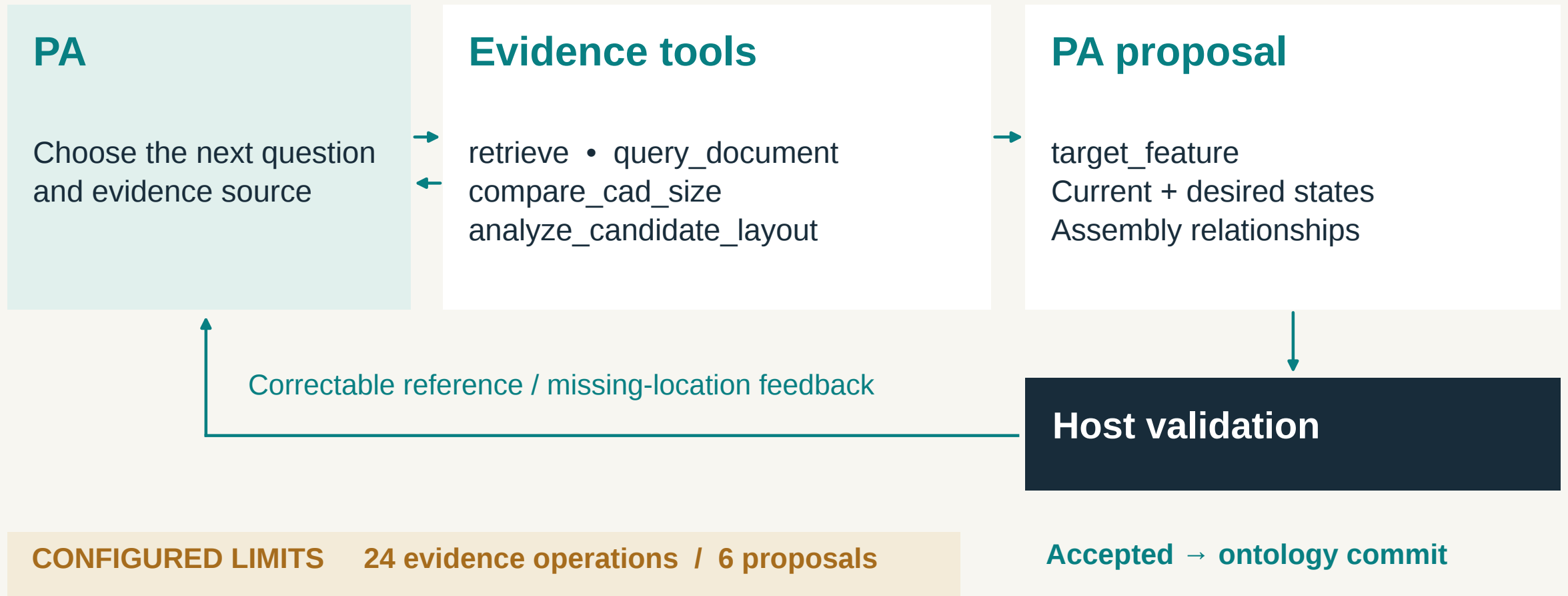
Goal + assignment

target_feature
Pairwise relationships
MoveIt arm checks

PA chooses the evidence order. Phase 4.3 contracts apply throughout.

HANDOFF TO PHASE 5 Selected RA context → primitive_steps → bounded validation

PA decides what evidence to request next



Documents, CAD and RGB-D contribute different facts

01 DOCUMENTS

Intended relationships

Approved pages + figures
PA-authored query
Citations + uncertainty

Supports what should be assembled.

02 CAD

Candidate geometry

Measured mesh dimensions
All-candidate size comparison
No tool-selected winner

Supports shape and size comparison.

03 RGB-D

Observed evidence

Neutral segments + morphology
Bounds, centroids and layout
Approved frame conversion

Supports where observed objects are.

PA connects part identity, task role and the desired assembly relationship.

The host checks contracts; PA owns meaning

PA authors

- Part identities and task roles
- Current and desired state meaning
- Pairwise assembly relationships
- Coverage of the user requirement

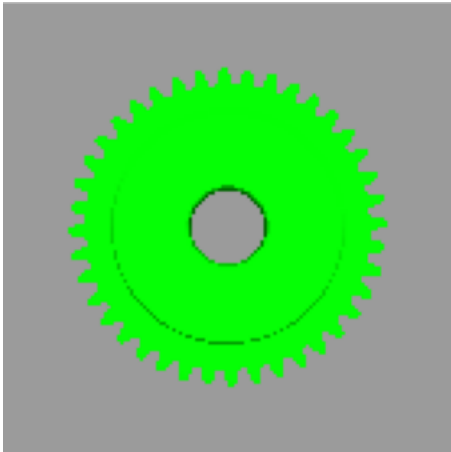
Host validates

- Schema + supplied PPR constraints
- Exact citations and typed bindings
- Hashes and source provenance
- Required current / desired locations

Semantic correctness needs separate evaluation. Source uncertainty stays visible.

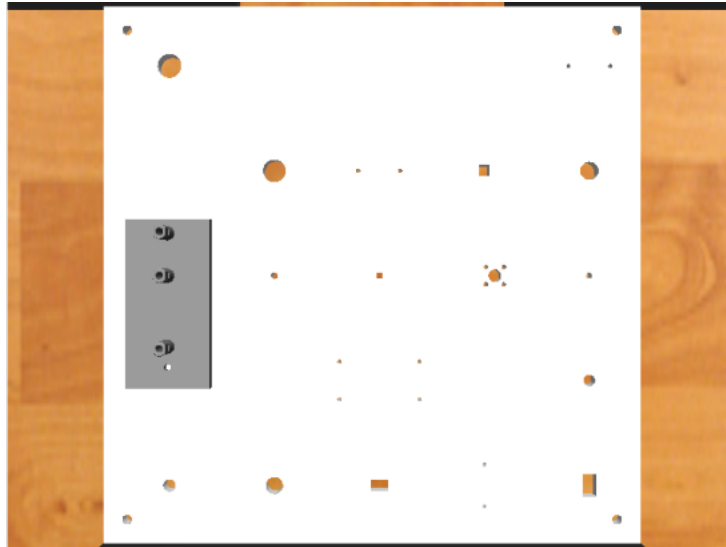
Saved example: “assemble medium gear.”

Observed candidate crop



PA binding: medium gear

Observed assembly workspace



PA'S SAVED INTERPRETATION

current_state

medium gear separate from
its intended middle gear shaft

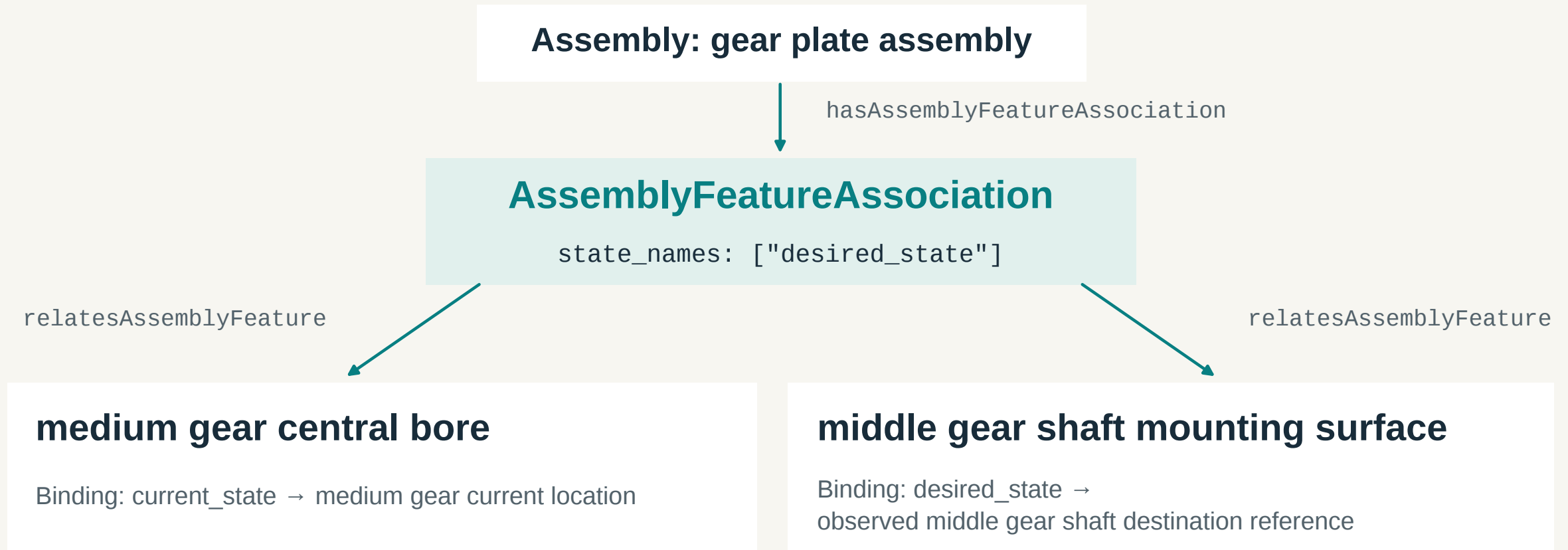
desired_state

medium gear mounted on the
middle gear shaft of the gear plate

The shaft provides an observed reference; a final gear pose is still unspecified.

Saved trace: 9 evidence operations • 1 proposal • Images are observed references; task correctness was not independently scored here.

One desired relationship links two grounded features



Association membership and endpoint observation bindings are independent.

PA checks every capable arm before selecting

Same evidence for both arms: 2 current references + 1 desired reference

Arm	All bound positions	Mean distance*
xarm6	accepted	0.549 m
ur5e	accepted	0.627 m

*Mean straight-line distance from the live TF base to the grounded current_state locations.

Movel **GetMotionPlan**

Live robot state + scene
Configured group / TCP
5 mm position tolerance
Unconstrained tool orientation

PA selected xarm6 • Accepted position plans are required; distance provides an advisory preference.

Completion hands grounded evidence to Phase 5

Grounded product

target_feature
Current + desired states
Pairwise relationships

Arm assignment

Selected resource
All-arm planning evidence
Assignment assertions

Traceable completion

Sources + typed records
Hashes + context snapshot
Persistent uncertainty

```
validation_scope:  
  moveit_state_location_reachability  
motion_validation_performed: true  
motion_executed: false
```

PHASE 5 CONTINUES WITH

Selected RA context → primitive_steps
Target calculations + modeled validation

Assembly success requires execution and independent outcome evidence.